

Multi-Stock S&P 500 Direction Prediction with Systematic Feature Selection

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1 Introduction

Applying sophisticated machine learning to financial markets presents a paradox. While advanced architectures like LSTMs and Transformers excel at capturing complex temporal patterns, their performance on next-day stock direction prediction is empirically capped at 52–60% accuracy, which is barely outperforming random chance (Fischer and Krauss, 2018). This ceiling is not an algorithmic limitation but a fundamental data problem: daily market data is dominated by an extremely low signal-to-noise ratio (Taleb, 2001).

To navigate this high-noise environment, two research streams have shown promise. The first is multi-scale feature engineering, which creates features across various time horizons (e.g., 5-day to 200-day) to capture a richer set of market dynamics (Wang et al., 2024). The second is systematic feature selection, which prunes large feature sets to isolate predictive signals and mitigate overfitting from noise (Wong and Barahona, 2023). While impactful, these approaches have largely been pursued in isolation. This study bridges that gap, investigating a central research question: Can systematic feature selection, when applied to a rich set of multi-scale engineered features, enhance the performance and generalization of stock direction prediction models?

To answer this, we conduct a comparative analysis of three models representing different complexity classes—Logistic Regression, Random Forest, and the state-of-the-art FT-Transformer. We evaluate their performance on a full 128-feature set, a baseline 5-feature set, and a 20-feature subset selected via Random Forest Mean Decrease in Accuracy. The analysis is performed on daily OHLCV data for 15 diverse S&P 500 stocks from January 2022 to the present (Yahoo Finance, 2025).

2 Literature Review

2.1 The Tension Between Market Efficiency and Empirical Prediction

The Efficient Market Hypothesis (Fama, 1970) posits that historical price data cannot be used to generate consistent, abnormal profits. Yet, a growing body of research demonstrates that machine learning models can uncover statistically significant predictive patterns (Krauss et al., 2017). A common approach is to use technical indicators as features—quantifiable metrics for market momentum, volatility, and trend, such as RSI and MACD, which have been formalized in foundational texts (Murphy, 1999) and validated in modern algorithmic trading. Our work operates within this paradigm, using a large set of such indicators as the basis for our multi-scale feature engineering.

2.2 Methodological Rigor in Financial Forecasting

Two critical methodological challenges in financial ML are data leakage and spurious correlations. To prevent lookahead bias, where future information contaminates the training process, all predictive features must be properly lagged, a principle emphasized by López de Prado (2018) as fundamental to achieving valid out-of-sample performance. Furthermore, financial data often contains persistent seasonal patterns, or calendar anomalies, such as day-of-the-week effects (Smith et al., 2024; Hasan and Al-Najjar, 2024). While their predictive power may have diminished in modern markets, these anomalies can encode structural information about institutional behavior. This study incorporates both principles: rigorous temporal lagging of all features and the inclusion of calendar effects to build a methodologically sound baseline.

2.3 Advancements in Tabular Data Modelling: FT-Transformer

While tree-based ensembles like Random Forest are traditionally strong performers on tabular data, recent advancements have adapted the Transformer architecture for this domain. Gorishniy et al. (2021) introduced the FT-Transformer, which treats each feature as a “token” and uses a self-attention mechanism to learn complex inter-dependencies. Unlike models that treat features independently, FT-Transformer creates contextualized feature representations, achieving performance competitive with state-of-the-art methods. Its inclusion in our study allows us to test our research question against a cutting-edge deep learning architecture specifically designed for the type of high-dimensional feature matrix we have constructed.

3 Methods

3.1. Data and Feature Engineering

The dataset was constructed from daily OHLCV (Open, High, Low, Close, Volume) data for 15 S&P 500 stocks, sourced from Yahoo Finance for the post-pandemic period (January 2022–present). After applying a forward-fill to handle rare missing values, the predictive target was defined to prevent lookahead bias. The target, *Direction*, is a binary variable set to 1 if the next day’s closing price is higher than the current day’s ($\text{Close}_{t+1} > \text{Close}_t$), and 0 otherwise. In total, the panel comprises 11,220 stock–day observations spanning 949 trading days post-COVID and covering 15 leading S&P 500 constituents from various sectors.

A comprehensive set of over 100 predictive features was engineered for each stock using a multi-scale approach. This involved calculating technical indicators (e.g., RSI, MACD), volatility metrics, and momentum signals across multiple rolling windows, ranging from short-term (5-day) to long-term (200-day). To ensure temporal integrity, all engineered features were lagged by one period, guaranteeing that predictions for time $t + 1$ use only information available at time t .

3.2. Systematic Feature Selection

To investigate our research question, a reduced 20-feature subset was created using Random Forest Mean Decrease in Accuracy (RF-MDA). This method was chosen over simpler alternatives for two key reasons. First, its permutation-based importance metric effectively captures non-linear relationships and feature interactions. Second, it is robust to the multicollinearity that is pervasive in financial datasets where many technical indicators are inherently correlated. In contrast, alternatives like Lasso regression were rejected due to their inherent bias towards linear relationships, while dimensionality reduction techniques like PCA were avoided because they sacrifice the interpretability of the original features.

3.4. Algorithms

Three models of varying complexity were selected to systematically evaluate the impact of feature selection across different architectural paradigms: linear, ensemble, and deep learning approaches.

Logistic Regression serves as an interpretable linear baseline, establishing whether directional prediction can be achieved through linear combinations of features. Its simplicity allows assessment of whether non-linear modeling is necessary.

Random Forest captures non-linear feature interactions through ensemble learning. Beyond prediction, it provides the RF-MDA importance scores used for feature selection, making it central to our methodology.

FT-Transformer represents state-of-the-art deep learning for tabular data (Gorishniy et al., 2021). The architecture treats each feature as a token: a Feature Tokenizer embeds continuous features into vector space, a learnable CLS token aggregates global information, and stacked Transformer blocks use self-attention to learn feature interactions. The CLS token output feeds into an MLP head for binary classification. This model establishes a performance ceiling for complex feature interactions.

3.5. Model Evaluation and Hyperparameter Tuning

Model generalisation was estimated using a nested walk-forward cross-validation protocol that respects temporal structure. The outer loop consists of 10 chronological folds with an expanding training window, simulating realistic periodic model retraining scenarios. Each outer fold divides the data into a growing training set and a subsequent test set. Hyperparameter tuning was integrated into the inner loop of this nested structure. For each of the 10 outer folds, a 3-fold inner walk-forward cross-validation identified optimal hyperparameters from predefined grid spaces, where specific hyperparameter spaces explored were: Logistic Regression: *C* (inverse regularisation strength) values of [0.1, 1.0, 10.0]. Random Forest: *n_estimators* [100, 200, 300] and *max_depth* [10, 20, 30]. FT-Transformer: *learning_rate* [0.01, 0.001, 0.0001] and *embed_dim* [16, 32, 64].

The best-performing parameter combination from the inner loop was then used to train the model on the full outer training set for evaluation on the corresponding outer test fold. The final cross-validation performance was computed by averaging metrics across all 10 outer folds. To determine the optimal hyperparameter configuration for the final model, we selected the parameter combination that was most frequently chosen across the 10 outer folds. This final model was then trained on the complete training dataset and evaluated on a separate hold-out test set to assess ultimate generalisation performance. All cross-validation and hyperparameter tuning loops, as well as the calculation of performance metrics, were implemented from scratch to ensure full compliance with the project requirements.

4 Results and Discussion

4.1. Performance Overview

Table 1 presents comprehensive performance metrics across all model-feature combinations evaluated through nested walk-forward cross-validation. Test accuracy ranges from 51.60% to 55.51%, with F1 scores between 60.50% and 71.39%. These results situate our findings within the established empirical bounds of financial prediction literature, where directional accuracy rarely exceeds 52–60% (Fischer and Krauss, 2018), fundamentally constrained by Taleb’s observation that daily financial markets contain approximately 95% noise and only 5% signal (Taleb, 2001).

Table 1: Nested walk-forward CV (mean $\pm$ std) vs. hold-out Test performance with optimal hyperparameters(HP).

Model	Features	CV Acc(%)	CV F1(%)	Test Acc(%)	Test F1(%)	Optimal HP
Logistic Regression	5	54.69 $\pm$ 3.64	70.52 $\pm$ 3.23	55.51	71.39	C=1
Logistic Regression	20	52.90 $\pm$ 3.78	66.55 $\pm$ 4.39	54.31	68.56	C=1
Logistic Regression	128	51.27 $\pm$ 3.11	62.19 $\pm$ 4.24	51.60	60.50	C=1
Random Forest	5	51.76 $\pm$ 2.02	61.63 $\pm$ 3.72	52.76	65.16	n=100, d=10
Random Forest	20	52.68 $\pm$ 3.13	63.63 $\pm$ 3.62	54.13	66.52	n=200, d=20
Random Forest	128	52.05 $\pm$ 2.96	61.34 $\pm$ 4.09	53.11	64.96	n=200, d=20
FT-Transformer	5	54.75 $\pm$ 3.74	70.53 $\pm$ 3.14	55.51	71.39	lr=1e ⁻³ , ed=32
FT-Transformer	20	54.82 $\pm$ 3.77	69.82 $\pm$ 4.22	55.51	71.39	lr=1e ⁻³ , ed=32

HP legend: C = inverse regularisation; n = #trees; d = max depth; lr = learning rate; ed = embedding dim.

4.2. Discussion

The Classification Collapse. Our central finding is that all models, regardless of architectural complexity or feature set size, ultimately failed. As illustrated by the confusion matrices (Figure 1), both Logistic Regression and the FT-Transformer learned to exclusively predict the majority class (“Up”), achieving a test accuracy of 55.51% that simply mirrors the class prevalence. The high F1-score of 71.39% is not an indicator of quality but a statistical artifact of this biased strategy, as the models show zero precision or recall for the minority class. This classification collapse is the direct result of an extremely low signal-to-noise ratio (est. 1:19), which makes distinguishing predictive patterns from random noise practically impossible at a daily frequency.

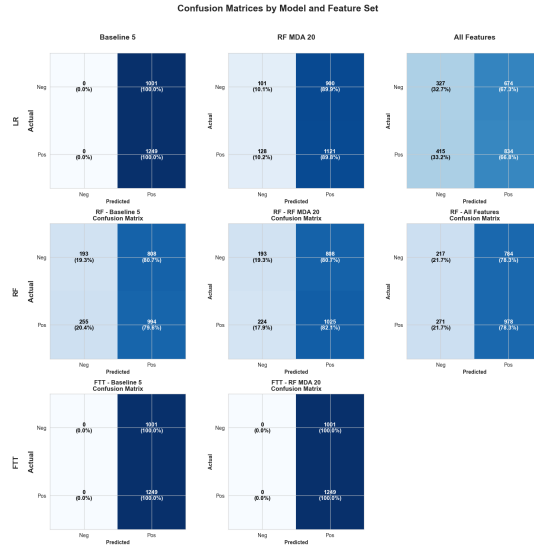


Figure 1: Confusion matrix on the **hold-out test set** across all models.

The Curse of Dimensionality in High-Noise Environments. Logistic Regression shows the sharpest drop—test accuracy decreases by 3.9% and F1 by 10.9%. This pattern persists even under nested and temporally aware cross-validation, confirming that the issue extends beyond conventional overfitting. In a market with a signal-to-noise ratio near 1:19, each additional feature is more likely to add noise than information. As dimensionality increases, the model capacity to memorize random fluctuations grows faster than its ability to extract true patterns. Specifically, Logistic Regression’s performance degraded sharply as dimensionality increased, proving stronger regularization is insufficient to counteract the added noise. The more resilient Random Forest also deteriorated, illustrating that the robustness of ensemble methods is finite in high-dimensional, low-signal environments.

Why FT-Transformer Fails to Deliver Expected Gains. The study’s most significant finding is that the complex FT-Transformer performed identically to a simple Logistic Regression, with both achieving 55.51% test accuracy and 71.39% test F1. This equivalency arose because both models defaulted to the same baseline strategy: predicting the majority class, which constitutes approximately 55% of the data. The Transformer’s architectural sophistication offered

no advantage due to three fundamental data limitations: 1) An extremely low signal-to-noise ratio (1:19) created a hard performance ceiling. 2) The effective sample size was likely insufficient for the complex model to learn generalizable patterns. 3) A dominant class imbalance skewed the learning process toward the majority class. This convergence suggests that, in line with the efficient-market hypothesis, strong, exploitable predictive signals may be absent at a daily frequency.

The Limits of Feature Selection. This experiment also highlights that in a low-signal regime, feature selection risks becoming a model-specific overfitting tool rather than a method for uncovering generalizable truths. The 20-feature subset selected by Random Forest MDA improved its parent Random Forest model (+1.37%) but degraded Logistic Regression (-1.20%). This demonstrates that RF-MDA did not find a universally predictive signal; instead, it identified a subset of noisy features that were idiosyncratically useful for the Random Forest’s tree-based structure. In high-noise data, even robust permutation-based methods are susceptible to identifying spurious correlations that fail to generalize across different model architectures.

Alternative Approaches Considered. Several alternative feature selection methods were considered but deemed unsuitable for our study. Recursive Feature Elimination (RFE) was computationally prohibitive within our nested cross-validation framework and demonstrated instability with highly correlated financial data. Principal Component Analysis (PCA) was rejected because it sacrifices the essential interpretability of original features and risks prioritizing noise over signal, as high variance in financial markets often represents noise. Similarly, lasso regularization’s implicit selection mechanism hinders the extraction of consistent feature subsets for the cross-model comparison central to our research. In contrast, RF-MDA was selected for its computational feasibility, its ability to produce explicit and comparable feature subsets, and its capacity to capture non-linear feature importance.

Practical Implications and Future Directions. The finding that models default to majority-class prediction in daily financial forecasting, driven by extreme class imbalance and low signal-to-noise ratios, has significant practical implications. This tendency renders accuracy a potentially misleading metric, as high performance may merely reflect baseline prediction rather than genuine pattern recognition. The convergence of both simple and complex models on this behavior underscores that architectural sophistication cannot overcome fundamental data limitations. Future research should therefore focus on either improving the signal-to-noise ratio—by extending prediction horizons or incorporating alternative data (e.g., sentiment, macroeconomic indicators)—or reframing the problem. Promising alternative approaches include applying class-balancing techniques, concentrating on rare high-confidence predictions, employing anomaly detection, or developing robust ensemble strategies.

5 Conclusion

In the high-noise setting of daily stock direction prediction, performance is constrained by the data’s low signal-to-noise ratio rather than algorithmic sophistication. Across all models—from Logistic Regression to the FT-Transformer—accuracy collapsed to ~55.51%, reflecting a systematic classification failure in which models learned to predict only the majority class. Architectural complexity thus offered no advantage, and systematic feature selection did not generalize: the RF-MDA subset aided its parent Random Forest but hurt Logistic Regression, indicating model-specific artifacts rather than robust signals. Progress will likely depend on improving the underlying signal—e.g., incorporating alternative data or extending prediction horizons—rather than further refining architectures alone.

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